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CS 340 Client/Server Development

Project Overview

For this project, I created an interactive dashboard for Grazioso Salvare that helps filter rescue animal data from the Austin Animal Center database. The dashboard allows the user to filter animals by rescue type, view matching records in a data table, see the selected animal's location on a map, and view breed distribution in a pie chart.

Required Functionality

My dashboard includes all required functionality for the project:

- Interactive data table
- Rescue type filter buttons for Water Rescue, Mountain/Wilderness Rescue, Disaster/Tracking, and Reset
- Geolocation map that updates when a row is selected
- Breed distribution pie chart that updates when filters are applied
- Grazioso Salvare logo and my unique identifier on the dashboard

Figure 1. Default/Reset Dashboard View

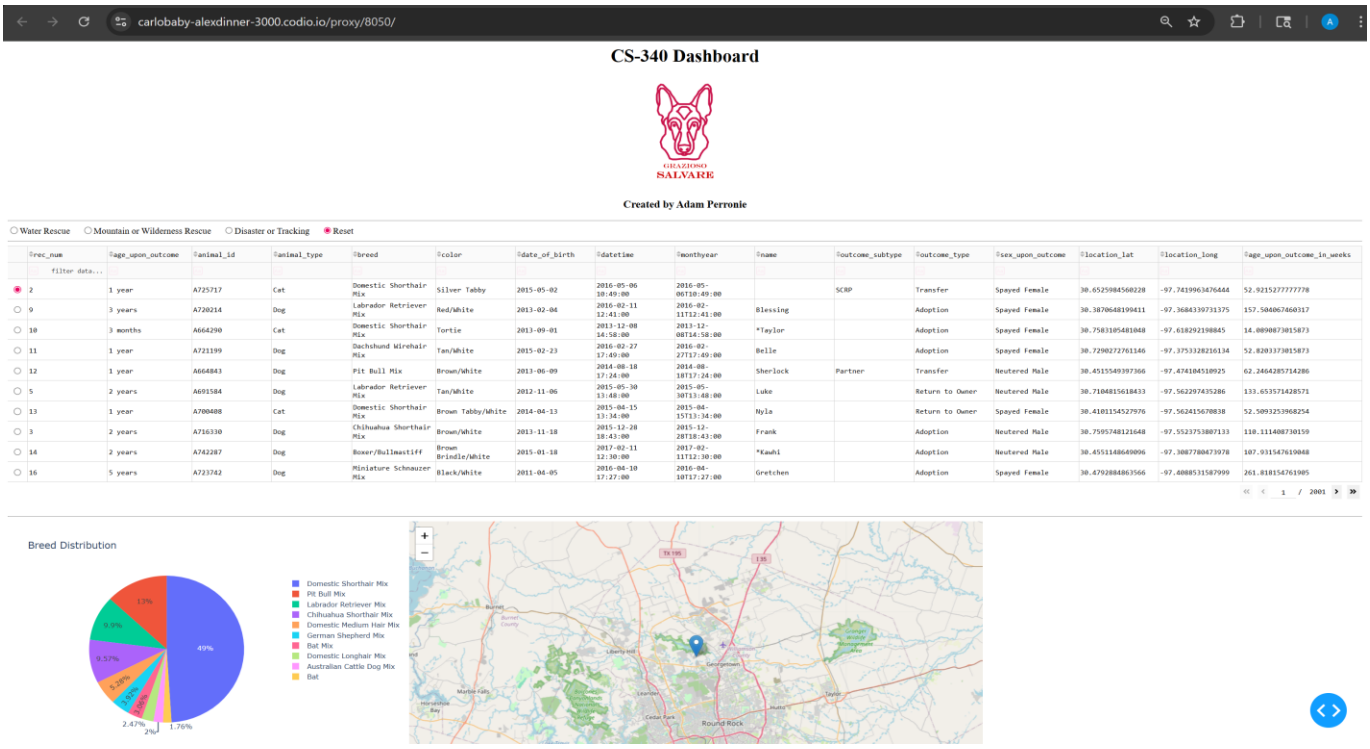


Figure 2. Water Rescue Filter Applied

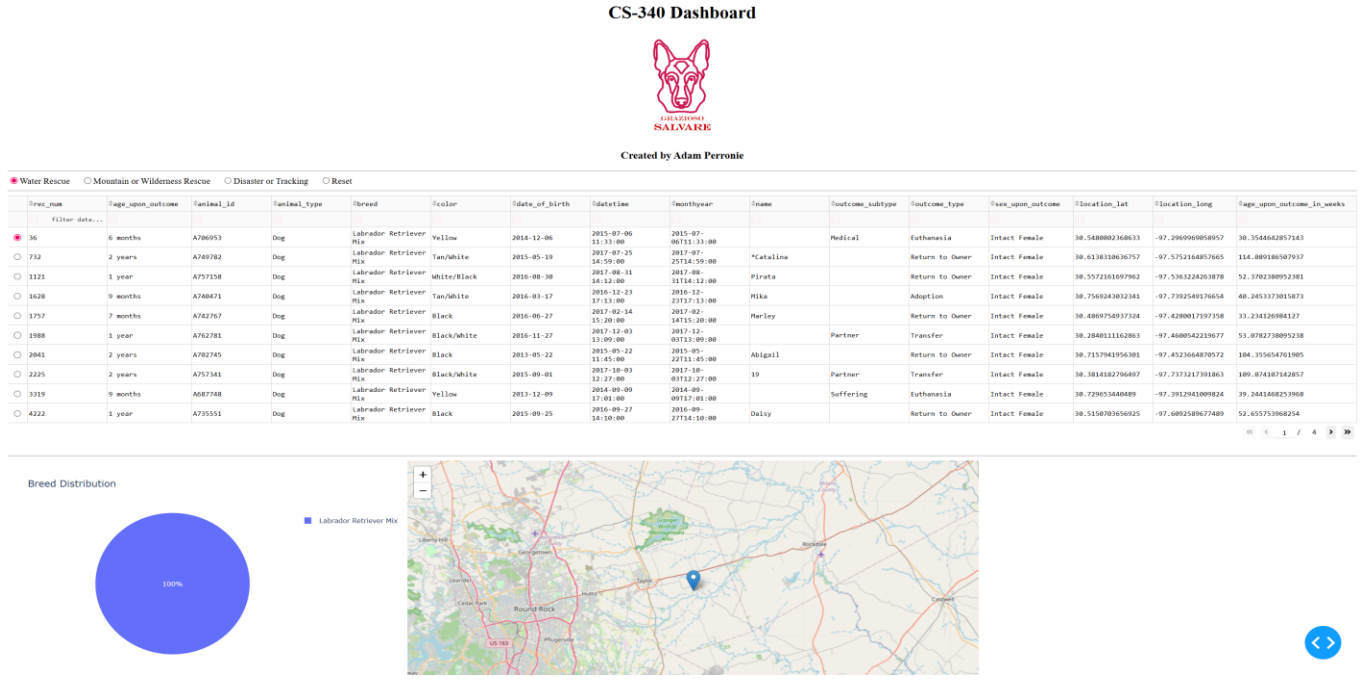


Figure 3. Mountain or Wilderness Rescue Filter Applied

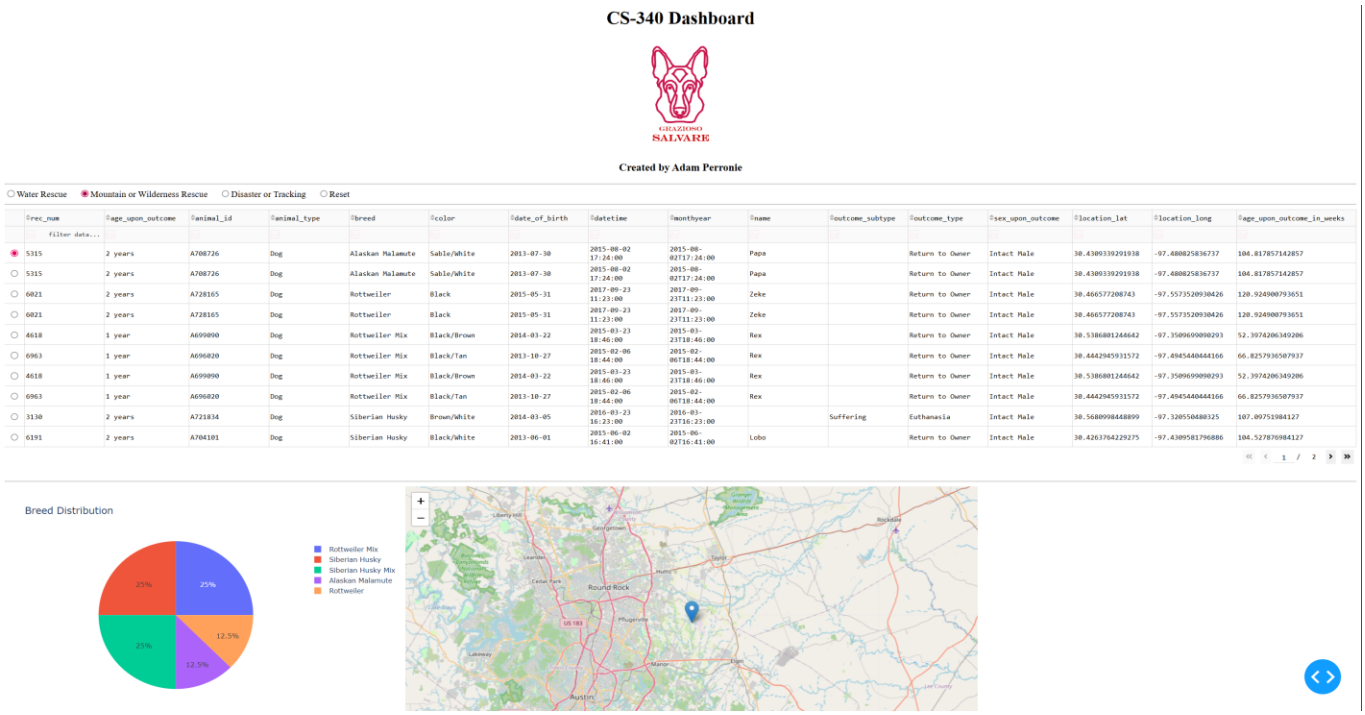
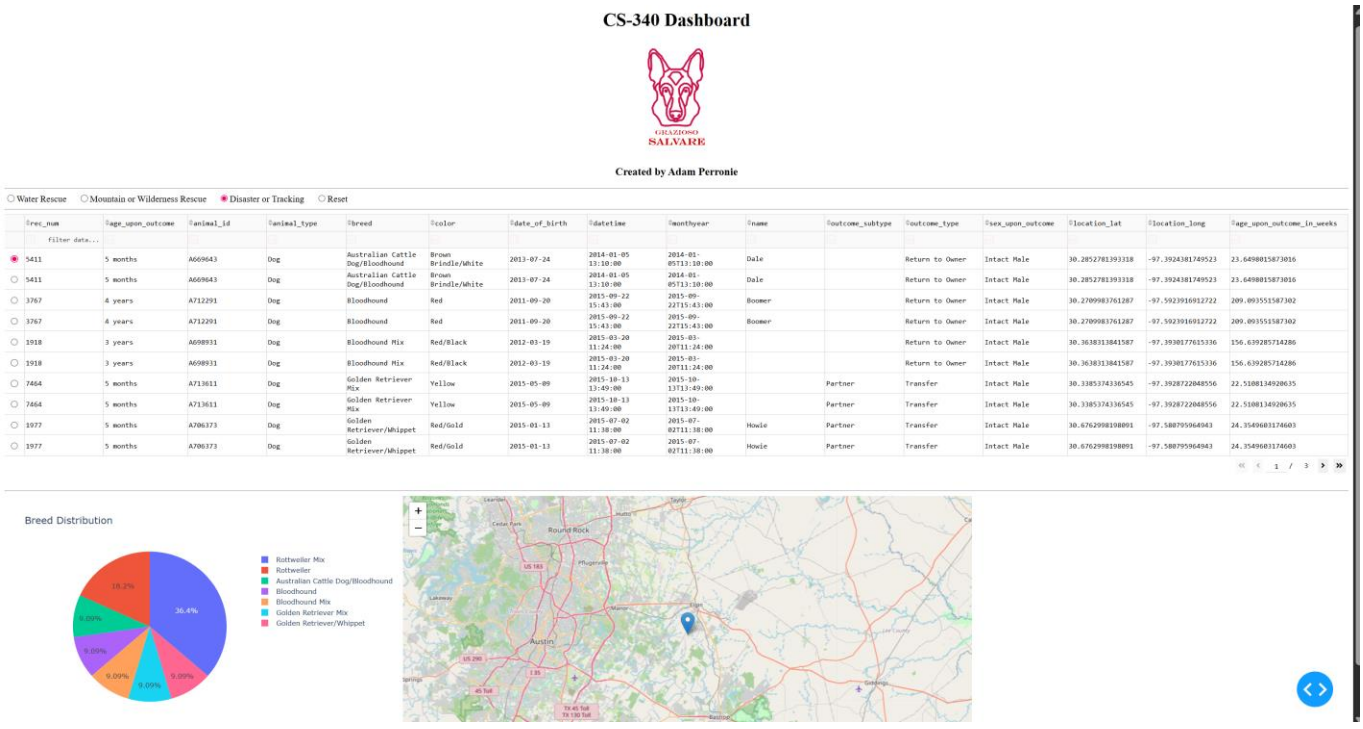


Figure 4. Disaster or Tracking Filter Applied



Tools Used and Why

MongoDB

I used MongoDB to store and retrieve the animal rescue data. MongoDB works well with Python and made it easy to query the data needed for the dashboard filters.

Dash

I used Dash to build the web dashboard and create the interactive layout of the application.

Plotly / Dash Leaflet

I used Plotly for the pie chart and Dash Leaflet for the geolocation map so the dashboard could visually display the filtered data.

Pandas

I used Pandas to organize and manipulate the data returned from MongoDB before displaying it in the dashboard.

Steps I Took to Complete the Project

1. Connected my CRUD Python module from Project One to the dashboard.
2. Added the Grazioso Salvare logo and my name to the dashboard layout.
3. Built the interactive data table.
4. Added radio button filters for each rescue type.
5. Created callback functions to update the data table when filters are selected.
6. Added a pie chart to show breed distribution.
7. Added a map that updates based on the selected animal.
8. Tested all filters and dashboard features to make sure they worked correctly.

Challenges and How I Solved Them

One challenge I had was getting the pie chart callback to work correctly with filtered data. I had to adjust the callback logic so the chart updated properly when the filters changed.

Another issue I ran into was making the map center correctly when a new animal was selected. I fixed this by updating the map callback so the center coordinates matched the selected row's latitude and longitude.

How to Reproduce the Project

To run this project:

1. Open the project in Codio/Jupyter Notebook
2. Make sure MongoDB is running and the Austin Animal Center dataset has already been loaded
3. Keep these files in the same folder:
    a. ProjectTwoDashboard.ipynb
    b. CRUD_Python_Module.py
    c. Grazioso Salvare Logo.png
4. Run all cells in the notebook
5. Open the dashboard link and interact with the filters, table, map, and chart

Final Summary

This project helped me practice connecting a database to a client-side dashboard and building an interactive web application using Python tools. The final dashboard gives Grazioso Salvare a way to filter and visualize rescue animal data in an easy-to-use format.